Deep Geospatio-Semantic Guided Network With Pseudo-Label Consistency for Domain-Adaptive Remote Sensing Segmentation

Jiawei Ning, Zhongle Ren, *Member, IEEE*, Biao Hou, *Senior Member, IEEE*, Weibin Li, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Domain-adaptive remote sensing image (RSI) semantic segmentation mitigates the overfitting problem that affects the effectiveness of segmentation, which results from the scarcity of high-quality labels and the cross-domain styles of ground objects. The effectiveness of domain-adaptive segmentation remains suboptimal in complex scenarios due to inadequate exploitation of latent geographic knowledge. Consequently, interclass ambiguity and boundary agnostic are further exacerbated under cross-domain transfer scenarios. To address this issue, we first devise a deep geospatio-semantic guided network named deep spatio-semantic awareness learning (DSSAL), which comprehensively investigates the potential spatial relationship and semantic correlation between classes of RSIs by geospatial-aware interaction and geosemantic-aware interaction, respectively. To mitigate class-wise cognitive deviation in the unlabeled domain, DSSAL domain adaptation (DSSAL-DA) is developed to further enhance the segmentation effect with the spatio-semantic domain alignment (SSDA) module in manifold cross-domain tasks. Furthermore, a pseudo-label consistency filter (PCF) is developed for DSSAL-DA to ensure reliability in self-training through cross-view consistency verification. Extensive experiments on two public datasets and a private dataset demonstrate the superiority of DSSAL and DSSAL-DA over the state-of-the-art (SOTA) methods for UDA semantic segmentation of RSIs.**

***Index Terms*— Domain adaptation, remote sensing images (RSIs), self-training, semantic segmentation.**

I. Introduction

LARGE-SCALE land use and land cover (LULC) classification serves as a cornerstone for strategic decision-making in wide fields such as urban planning [1], [2], precision agriculture [3], [4], and natural resource management [5], [6]. With the ever-increasing data scale of high-resolution remote sensing images (RSIs), the coverage of LULC mapping lags far behind the actual requirements. While deep learning-driven semantic segmentation has revolutionized cartographic processes through automated feature extraction, two critical bottlenecks persist.

1) *Annotations:* The exponential growth of high-resolution satellite imagery starkly contrasts with the scarcity of expert-curated annotations, causing fully supervised source-trained models to exhibit severe performance degradation when deployed in unseen target domains.
2) *Geo-Styles:* Multisensor discrepancies compounded by cross-regional heterogeneity induce substantial geo-styles, which have become an enormous challenge to extract and classify the features of ground objects from the unlabeled cross-geographical regions.

These challenges collectively hinder the development of robust feature representation learning frameworks capable of generalizing across diverse RSI scenarios. As shown in Fig. 1, the inherent limitations of fully supervised paradigm become particularly pronounced when addressing cross-domain challenges in cross-sensors, various landuse distributions, and diverse landcover styles. In particular, the domain disparities of different bands, diverse road construction, and distinct urban styles are intuitively demonstrated. In urban planning and remote sensing analysis, different sensor technologies exhibit unique spectral signatures when capturing the same terrain category, manifested through distinct color representations in imaging outputs [7]. This technological variation becomes particularly evident when examining infrastructural differences between urban and rural contexts, where road networks demonstrate substantial divergences in geometric characteristics, including pavement texture, carriageway width, and spatial configuration patterns [8], [9]. Furthermore, these disparities become compounded when considering terrain-dependent urbanization patterns, as slope gradients, hydrological conditions, and geological formations impose distinct constraints on urban morphology, thereby generating region-specific construction typologies. Currently, the state-of-the-art (SOTA) remote sensing foundation

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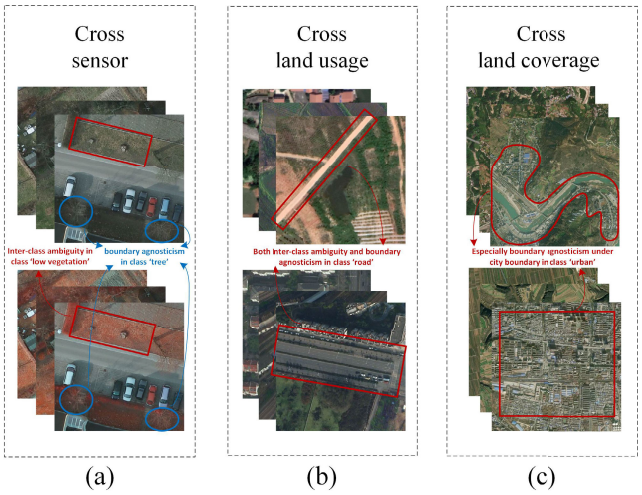


Fig. 1. Illustrations of the three classical cross-domain types in pairs. (a) Cross sensor in the ISPRS Potsdam dataset. (b) Cross land usage in the LoveDA dataset. (c) Cross land coverage in the Arcmap SX48 dataset. The upper and lower lines represent the source and the target domain, respectively. The domain discrepancies among different datasets reveal common interclass ambiguity and boundary agnosticism, which are highlighted by the red and blue boxes.

models [10], [11], [12], which are pretrained on petabyte-level datasets, exhibit constrained robustness due to label noise propagation and task-specific misalignment of spatial and semantic features. On the contrary, domain adaptation has emerged as an established mitigation strategy to bridge the annotation-efficiency divergence while enhancing model generalization. The early domain adaptation semantic segmentation method was deservedly inspired by the adaptation methods initially designed for image classification [13], [14], [15]. Contemporary approaches, such as [16], [17], [18], and [19], typically leverage joint source–target feature space alignment through adversarial discriminators or self-training with pseudo-label refinement, effectively capturing cross-domain invariant representations via deep neural architectures for semantic segmentation. At present, domain adaptation methods for RSI semantic segmentation (RSISS) mainly consist of adversarial training and self-training.

Specifically, in adversarial training, to adapt the model to the target domain, one or more discriminators are typically constructed to confound the features of the source and target domains. Domain shift predominantly manifests in the deeper layers of backbone networks, motivating the effectiveness of adversarial learning frameworks that leverage weight-shared intermediate features for cross-domain alignment [20]. Inspired by the work of [16] on natural images semantic segmentation, Chen et al. [21] utilized mutual information to measure and optimize the feature relationships between the source and target domains, thus improving the discriminability in remote sensing scenarios. Ma et al. [22] introduced cross-domain attention interaction in RSIs to distinguish and adapt to features from disparate domains. Additionally, the transferability of different feature layers in the discriminative network has been thoroughly explored in [20], and it has been demonstrated that cross-domain feature discrimination at shallow feature levels also plays a crucial role in the generalization of the source domain. These empirical analyses reveal that adversarial learning predominantly addresses domain-wise feature distribution shifts through global alignment strategies [23]. However, establishing class-specific semantic correspondence remains an outstanding challenge in pixel-wise prediction tasks, where intraclass variance and interclass similarity across domains significantly impede fine-grained class-wise adaptation. Consequently, apart from considering interdomain alignment, it is also necessary to prioritize interclass confusion and intraclass variance for different domains.

A straightforward approach is to generate elaborate pseudo-labels for the target domain [24]. With the development of self-supervised learning, the self-training strategy has been introduced to derive the intraclass invariance across domains, strengthen the fine-grained domain differences at the class level, and generate reliable pseudo-labels. Jiang et al. [25] resorted to the contrastive learning strategy of class prototypes to refine the pseudo-labels. In the past three years, Hoyer et al. [18], [19], [26] adopted methods such as rare class sampling, context-aware correlation, and mask consistency model to mitigate the local domain differences between the source and target domains, respectively. Although RSIs diverge greatly from natural images in terms of imaging, scenes, and other aspects, the abovementioned methods provide valuable insights for cross-domain remote sensing scenarios. Chen et al. [27] and Jin et al. [28] optimized the uncertain regions in the pseudo-labels by leveraging the posterior probability relationships between classes. Ma et al. [29], Peng et al. [30], and Zeng et al. [31] constructed different class-level prototype optimization strategies to separate cross-domain features. Ran et al. [32], Gao et al. [33], and Li et al. [34] achieved data augmentation through cross-domain image fusion, enabling the model to conduct joint learning across domains. The works mentioned above effectively enhance cross-domain segmentation through versatile general-purpose strategies. However, the pivotal challenges of the effective utilization of geographical knowledge in RSIs remain inadequately addressed, such as the spatial distribution of the cross-domain land cover class and the adjacent dependency relationships of land use class, leading to persistent performance challenges under domain shift conditions.

In this article, geospatio-semantic knowledge, comprehensively derived from the inherent class distribution and potential class relationship, is employed to guide the semantic segmentation of RSIs. Therefore, a novel deep geospatio-semantic-guided method named deep spatio-semantic awareness learning (DSSAL) is presented, which is designed to derive the rational class-wise context from the perspectives of geospatio-semantic awareness. Furthermore, in order to alleviate cross-domain discrepancy and guarantee reliability in self-supervised learning, DSSAL domain adaptation (DSSAL-DA) is developed based on DSSAL, including geospatio-semantic domain alignment (GeoSSDA) module and pseudo-label consistency filter (PCF) strategy. Specifically, DSSAL is composed of a dual-branch structure, namely the geospatial-awareness interaction (GeoSpaAI) and the geosemantic-awareness interaction (GeoSemAI). These two

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components discern the precise interclass distribution regularities and alleviate the pixel-level class semantic confusion. DSSAL-DA leverages the geospatio-semantic representation of DSSAL to align the cross-domain spatial distribution and interclass semantic relationship, which is incorporated into the pseudo-label generation process to facilitate reliable self-training.

Our method demonstrates significant class-wise integrity and boundary separability in RSISS and manifests commendable generalization for domian adaptation in disparate remote sensing. The main contributions of the proposed method can be summarized as follows.

1) To the best of our knowledge, the proposed DSSAL is a novel RSISS method in the context of the geographical relationship guidance.
2) A dual-branch (GeoSpaAI and GeoSemAI) interaction framework is developed, which serves as deep guidance to focus on the potential distribution of spatial and semantic features in RSIs.
3) In DSSAL-DA, GeoSSDA aligns cross-domain spatio-semantic features to further facilitate a thorough interaction of geographical knowledge, and in combination with consistent representations, a reliable pseudo-label optimizing strategy PCF is designed.

The rest of this article is organized as follows. Section II makes a description of the related UDA semantic segmentation works. Section III elaborates on the details of the proposed DSSAL and DSSAL-DA. Experimental setup and discussion about the experimental results have been demonstrated in Section IV. Finally, Section V concludes this article.

II. Related Work

A. UDA for RSISS

The diverse classes and feature distributions in real-world scenarios render fully supervised single-domain methods ineffective, while unsupervised domain adaptation (UDA) addresses the prevalent over-fitting that models often encounter in cross-domain scenarios [35]. Although the UDA methods for natural images' semantic segmentation have provided valuable technical experience, these tasks are basically oriented toward migration from the synthetic image domain to real-world scenarios, lacking consideration for the characteristics of RSIs. Moreover, RSIs present inherent domain adaptation challenges stemming from their distinct scale variations, diverse scene characteristics, and notable style discrepancies [36], which impose formidable constraints on the optimization of model performance. In recent years, this situation has instigated substantial research initiatives. With the aim of achieving a balance in domain adaptation ability, certain studies employ a multistage strategy to execute this task. Zhang et al. [37] and Huang et al. [38] employed a progressive adaptation framework of easy-to-adapt and then hard-to-adapt to alleviate the complex disparate cross-domains. To enhance the model's perception in label agnostic domains, data augmentation stands as an effective and efficient approach. Some studies implement the data-mixing method, whereas others utilize the class-balanced sampling method. In [39], a category-level assignment problem with additional constraints was employed to model the category-level relationship between the source and target domains. Ran et al. [32] proposed a hybrid training approach that combined masked regions with the source and target domains. Chen et al. [27] utilized various data augmentation techniques to simulate common domain shift phenomena observed in RSIs. Li et al. [34] designed a mask enhancement and balanced sampling that compelled the model to infer the categories of the masked areas using contextual information and increased the sampling probability of small-scale and few-shot classes, respectively. In addition, in an attempt to align cross-domain features, several studies have employed comprehensive joint strategy solutions. In [23], contrastive learning and adversarial learning were considered jointly to represent space and spatial structure in an end-to-end model. It cascaded aligned source and target domains in terms of feature representations and spatial structures. Li et al. [40] proposed a semantic-preserved generative adversarial network with ClassMix [41] that carried out unbiased translation from the source-to-target domain and further managed to align the spectral information or even the imaging mode. Ma et al. [22] employed a boosted attention module with multidiscriminators to align cross-domain feature maps while enhancing domain-general features. Recently, self-training produces pseudo-labels by exploiting the unlabeled data in the target domain. It attains progressive domain adaptation in a cost-effective manner and manifests powerful practical value in RSISS.

B. Self-Training for UDA

The essence of UDA self-training lies in generating reliable pseudo-labels for the target domain while operating under the constraint of aligning the domain features [24]. Typically, the approaches of establishing a threshold for pixel-level probabilities are employed to eliminate uncertain predictions and prevent significant negative transfer in the context of self-supervised learning. Chen et al. [27], Li et al. [34], and Liu et al. [42] were jointly implemented in the context of the RSISS, in conjunction with a specifically set threshold and the domain alignment strategies. Specifically, Liu et al. [42] proposed a multilayer pseudo-label correction strategy that incorporates thresholds for the purpose of correcting pseudo-labels at the class, pixel, and superpixel levels. Owing to the substantial diversity in cross-domain disparities, fixed thresholds prove ineffective when it comes to filtering out ambiguous categories. In response, existing research [28], [43] had devised the dynamic threshold approach to mitigate the overfitting of the source domain stemming from various cross-domain challenges. Jin et al. [28] presented a dynamic and adaptive self-training approach that integrates dynamic pseudo-label sampling, distribution matching, and adaptive threshold updating. This approach alleviated the issue of class distribution disparity and guaranteed alignment with the ground-truth (GT) distribution. He et al. [43] employed a dynamic loss correction, which mitigated domain disparities by dynamically modeling the noise distribution of pseudo-labels in the target domain, leveraging a noise transfer matrix. Inspired by the prototypical network [44] in

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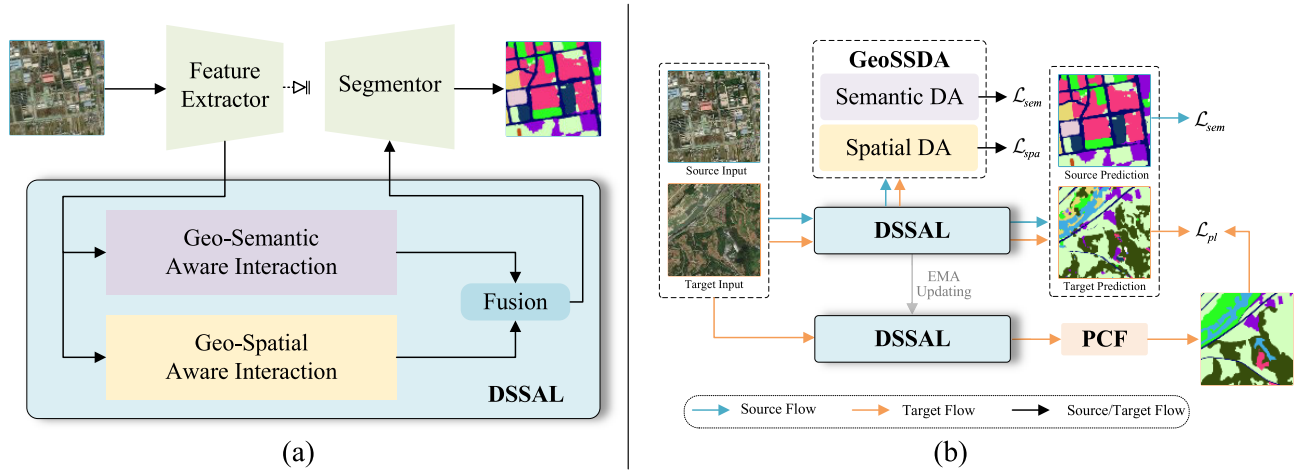


Fig. 2. Framework of the proposed DSSAL and DSSAL-DA. (a) Semantic segmentation method DSSAL in RSIs with geographical correlation guidance. (b) DSSAL is applied to cross-domain remote sensing scenarios and a domain-adaptive RSISS network DSSAL-DA is represented.

few-shot learning, class-level prototypes are constructed to guide the model to perceive the centroid of cross-domain features, thus enabling the model to conduct self-training in the target domain effortlessly. In UDA researches of RSISS [29], [30], [31], [33], [45], [46], [47], [48], the prototype fitting of classes places emphasis on delving into the category features with intricate semantic content. Zhang et al. [45] captured the commonalities and differences between domains and gradually optimized the feature representation through prototype-guided domain-aware progressive representation learning. Gao et al. [33] combined prototype learning with context enhancement. By constructing and utilizing prototypes, it strengthened the contextual information and improved the model's segmentation ability in cross-domains. Ma et al. [29] conducted contrastive learning based on prototypes, which prompted the model to learn more discriminative features. Zhu et al. [46] utilized the invariant domain-level prototype memory to store domain-invariant features, which aided the model in performing accurate segmentation. Zhao et al. [47] combined prototype learning and employed a self-training guided decoupling adaptation method to generate pseudo-labels, effectively decoupling the features of different domains. The work [48] was based on a contrastive adaptation method inspired by causal prototypes, which utilized causal prototypes to guide contrastive learning and optimized the model's adaptability to different domains. Peng et al. [30] leveraged prototype guidance for pseudo-label generation and combined perturbation-based active learning to select valuable samples. Zeng et al. [31] meticulously constructed and effectively utilized a prototype-guided paradigm to aid the model in more proficiently adapting to the feature distributions across diverse domains. However, these approaches demonstrate a deficiency in performance regarding class consistency and boundary integrity, especially during the semantic segmentation in complex cross-domain scenarios.

III. Proposed Method

A. Framework Overview

In this article, we first propose a semantic segmentation method DSSAL in RSIs with geographical correlation guidance, as shown in Fig. 2(a). Furthermore, DSSAL is applied to cross-domain remote sensing scenarios and a domain-adaptive RSISS network DSSAL-DA is represented as shown in Fig. 2(b). Specifically, we systematically investigate the latent geospatial patterns embedded in RSISS. To this end, we develop a dual-branch architecture, including GeoSpaAI and GeoSemAI, which enhances conventional backbone networks by providing them with cross-domain generalization for semantic segmentation. For domain adaptation, we design a GeoSSDA module to ensure cross-domain invariance in geospatio-semantic feature representation. Moreover, a pseudo-label generating strategy PCF is devised to enforce multiperspective consistency during self-training, thereby mitigating inaccuracies from low-confidence predictions. Implementation of the proposed methods is elaborated in Sections III-A–III-D.

B. Deep Semantic Spatial Awareness Learning

As shown in Fig. 3, we develop the dual-branch DSSAL framework, which leverages cross-domain transfer learning of class-wise geospatial interrelation and class-wise geosemantic correlation. It enhances the perception of backbone networks to internalize domain-specific geographical expertise through two core components, namely GeoSpaAI and GeoSemAI, which carry out vivid depiction of the land feature classes from the perspective of spatial relationship and semantic description, respectively.

1) Geo-Spatial Awareness Interaction: The GeoSpaAI is embodied in two aspects, including the global intraclass spatial distribution and the local interclass adjacent relationship, which flexibly characterize the geospatial features of RSIs. First, for the purpose of effectively fusing cross-scale spatial features, the deep features are carefully upsampled to the same spatial dimension as the shallow features. It can be formulated as

$$F_{\text{spa_init}} = \text{Concat}\big(\big[F_{s1}, \text{Up}(F_{s2}, F_{s3}, F_{s4})\big]\big). \quad (1)$$

To this end, the fused features $F_{\text{spa_init}}$ serve as the foundation for the geographical spatial perception interaction. Then,

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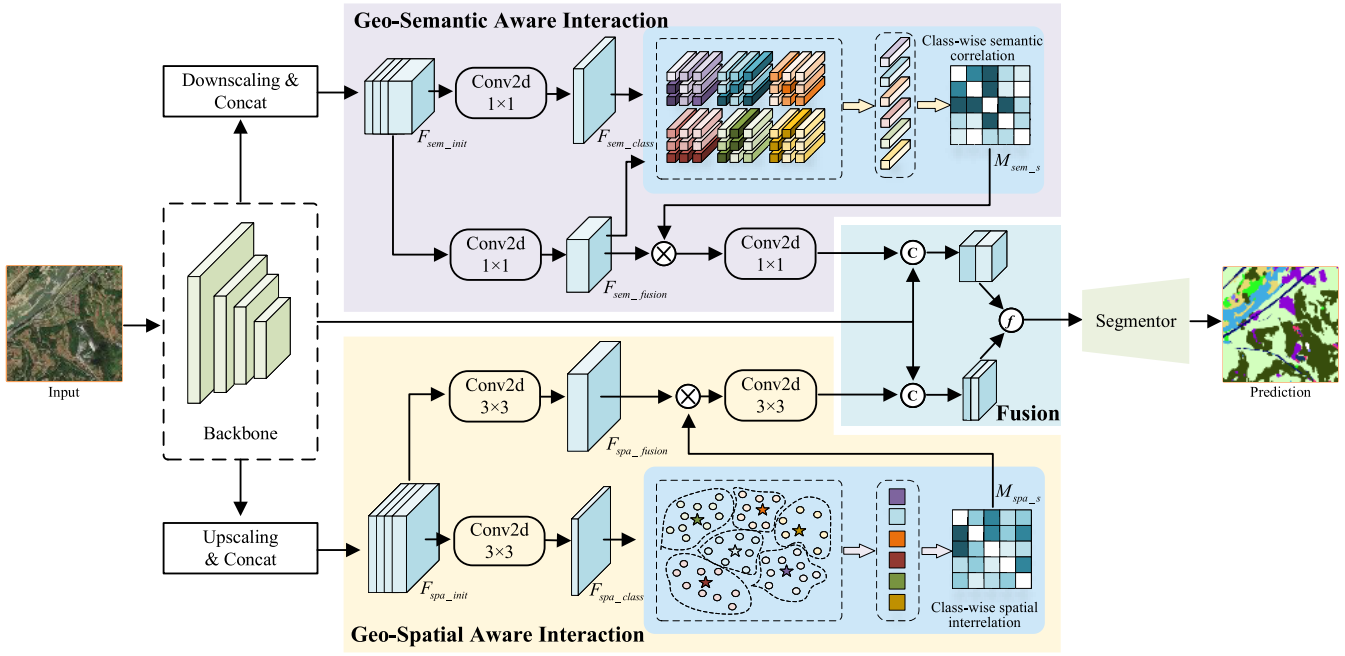


Fig. 3. Detailed framework of the proposed DSSAL.

$F_{\text{spa_init}}$ is subjected to a 3 × 3 convolution $\mathcal{F}_{\text{Conv2d}_{3\times3}}()$ to embed the spatial interrelation. It can be expressed as

$$P_{\text{spa_pro}} = \delta\left(\mathcal{F}_{\text{Conv2d}3\times3}\left(F_{\text{spa_init}}\right)\right) \tag{2}$$

$$P_{\text{spa_cls}} = \arg\max\left(P_{\text{spa_pro}}\right) \tag{3}$$

where δ is represented as Softmax() function and $P_{\text{spa_pro}}$ and $P_{\text{spa_cls}}$ adequately represent the class-probability distribution at the feature-level and class distribution in the prediction, respectively. Then, the geospatial interrelation is intuitively characterized by estimating the intraclass spatial distribution and the interclass adjacency relationship, as shown in Fig. 4(a).

Specifically, the intraclass spatial distribution is described by the class-connected component N_c

$$N_c = \mathcal{N}\left(I_{\text{spa_class}}\right) \tag{4}$$

where $\mathcal{N}(\cdot)$ represents the total summation of connected components in the cth class. N_c, as a regional-level statistical quantity for the category, effectively depicts the distribution of the class in the RSI.

Meanwhile, the interclass adjacent relationship is computed based on the count of class-wise adjacent pixels $A_{i,j}$ at the boundaries of the class-connected components

$$A_{i,j} = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1}\left[I_{\text{spa_class}}(h,w) = i \wedge \mathcal{K}(h,w,j) > 0\right] \tag{5}$$

where $A_{i,j}$ represents the number of adjacent pixels between the ith class and the jth class in the patch of RSI and $\mathcal{K}(\cdot)$ represents the fixed 3 × 3 convolution kernel [(1, 1, 1), (1, 0, 1), and (1, 1, 1)] used for computing the class relationships and $[\cdot]$ is an Iverson bracket, which assumes the value of 1 when the condition inside the bracket is true and 0 otherwise. Hence, the interclass pixel-level adjacent matrix is represented as

$$R_{i,j} = \frac{A_{i,j}}{N_c}. \tag{6}$$

To describe the spatial relationship precisely, $R_{i,j}$ is max–min normalized to a range of 0–1 and is utilized to illustrate each class-wise spatial correlations, which can be formulated as

$$\hat{R}_{\text{spa}} = R_{i,j} \cdot \max\left(\sum_{c=0}^{C-1} R_{i,c}, \epsilon\right)^{-1} \tag{7}$$

where $\hat{R}_{\text{spa}}$ represents the normalized interclass interrelation $R_{i,j}$ and ϵ represents the smoothing factor.

To mitigate the negative constraints of uncertain samples, $P_{\text{spa_class}}$ is assigned weights within the correlation matrix to precisely derive the interclass geospatial interrelation, which is composed of two steps as follows:

$$w_{\text{spa}} = \frac{1}{C}\sum \arg\max_{c\in C} P^{c}_{\text{spa_class}} \tag{8}$$

$$M_{\text{spa}} = w_{\text{spa}} \cdot \hat{R}_{\text{spa}} \tag{9}$$

where M_{spa} represents the interclass geospatial interrelation matrix. It guides the interclass spatial interaction of $F_{\text{spa_fusion}}$, which is reduced in dimension through a 3 × 3 convolutional layer

$$F_{\text{spa_fusion}} = \sigma\left(\mathcal{F}_{\text{Conv2d}_{3\times3}}\left(F_{\text{spa_init}}\right).\right. \tag{10}$$

Subsequently, the feature $F_{\text{seg_spa}}$, which obtains abundant geospatial interrelation, is extended via another 3 × 3 convolutional layer from $F_{\text{spa_fusion}}$. It can be defined as

$$F_{\text{seg_spa}} = \mathcal{F}_{\text{Conv2d}_{3\times3}}\left(F_{\text{spa_fusion}} * M_{\text{spa}}\right)\big) \tag{11}$$

where $F_{\text{seg_spa}}$ represents the geospatial feature of GeoSpaAI, which is finally input into the segmentor.

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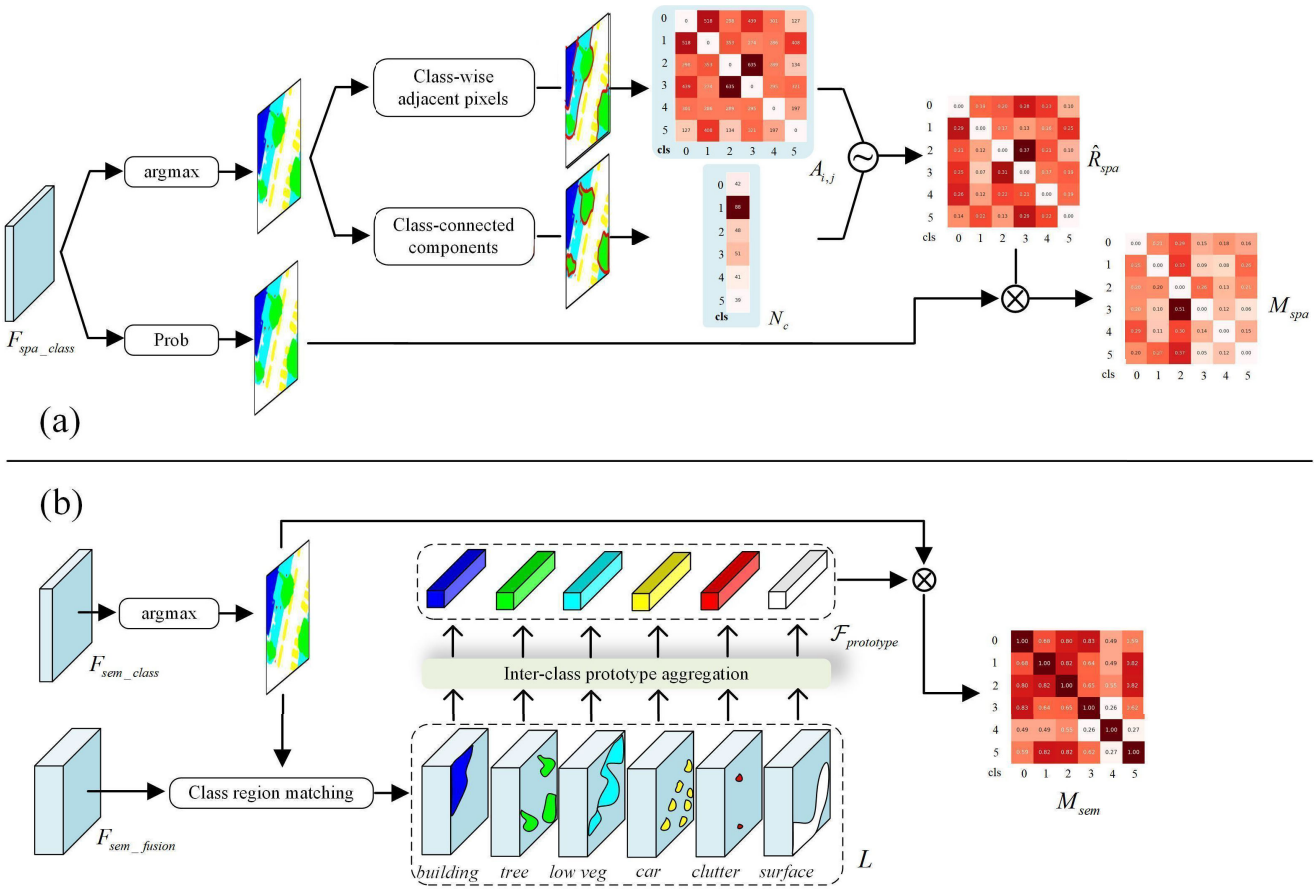


Fig. 4. Detailed illustrations of the relation matrices of GeoSpaAI and GeoSemAI. (a) Class-wise spatial interrelation. (b) Class-wise semantic correlation.

2) Geo-Semantic Aware Interaction: In this section, we have exhaustively investigated the potential of mining semantic correlation among land cover classes. Similar to GeoSpaAI, the features at various stages of the backbone network are integrated to obtain the coupled semantic features. Conversely, with the exception of the features from the final stage, the remaining features are downsampled to enable more effective extraction of geosemantic context

$$F_{\text{sem_init}} = \text{Concat}([F_{s4}, \text{Down}(F_{s1}, F_{s2}, F_{s3})]). \tag{12}$$

Then, two independent 1×1 convolutional layers are employed to aggregate the greenpixel-level semantic correlation and reduce the dimensionality, which effectively yields the cross-scale class representation and the aggregative semantic feature, respectively. They can be formulated as follows:

$$F_{\text{sem_class}} = \mathcal{F}_{\text{Conv2}d_{1\times1}}\left(F_{\text{sem_init}}\right) \tag{13}$$

$$F_{\text{sem_fusion}} = \mathcal{F}_{\text{Conv2}d_{1\times1}}\left(F_{\text{sem_init}}\right) \tag{14}$$

where $F_{\text{sem_class}}$ and $F_{\text{sem_fusion}}$ are jointly leveraged to extract the feature vectors, which collectively construct the class-wise semantic set L_c

$$L_c = \underset{c\in(0,\ldots,C-1)}{\text{list}} \left\{ \sum_{(h,w)\in(H,W)} F^{(h,w,c)}_{\text{sem_fusion}} \odot \mathcal{M}\left(F_{\text{sem_class}}\right) \right\} \tag{15}$$

where $\mathcal{M}(\cdot)$ represents the class-level one-hot operation for each pixel and $\odot$ denotes the dot product between semantic fusion feature $F_{\text{sem_fusion}}$ and dense position $\mathcal{M}(F_{\text{sem_class}})$ of each class c. Hence, L_c represents the aggregated feature of the cth class, calculated by summing up the features for each pixel belonging to the cth class.

The class prototypical representation is derived by interclass prototype aggregation, which calculates the mean vector of each class within set L_c, as depicted in Fig. 4(b). It effectively encapsulates the comprehensive geosemantic characteristics inherent to distinct geographical classes

$$\mathcal{F}^{c}_{\text{prototype}} = \left\{\overline{L_0}, \overline{L_1}, \ldots, \overline{L_{C-1}}\right\}. \tag{16}$$

Furthermore, a semantic correlation matrix is constructed through a covariance operation between each class in the prototype space $\mathcal{F}_{\text{prototype}}$, which is utilized to provide guidance for the feature extraction of the deep geosemantic correlation $\hat{R}_{\text{sem}}$

$$\hat{R}_{\text{sem}} = \frac{1}{n-1}\sum_{j=1}^{n}\left(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i\right)\left(\mathbf{x}_{kj} - \bar{\mathbf{x}}_k\right) \tag{17}$$

where n represents the dimension of semantic vector and $\mathbf{x}_i$ and $\mathbf{x}_k$ belong to $\mathcal{F}_{\text{prototype}}$. Analogous to GeoSpaAI, deep probability weighting is employed to prevent negative learning effects on the class-wise semantic representation

$$M_{\text{sem}} = w_{\text{sem}} \cdot \hat{R}_{\text{sem}} \tag{18}$$

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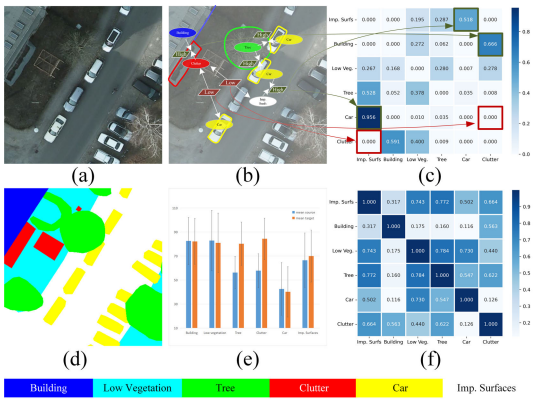


Fig. 5. Illustrations of the example of M_{spa} and M_{sem} in GeoSpaAI and GeoSemAI, respectively. (a) RSI. (b) Interclass adjacent diagram. (c) Geospatial interrelation matrix. (d) GT of RSI. (e) Class-wise feature distribution of RSI. (f) Geosemantic correlation matrix.

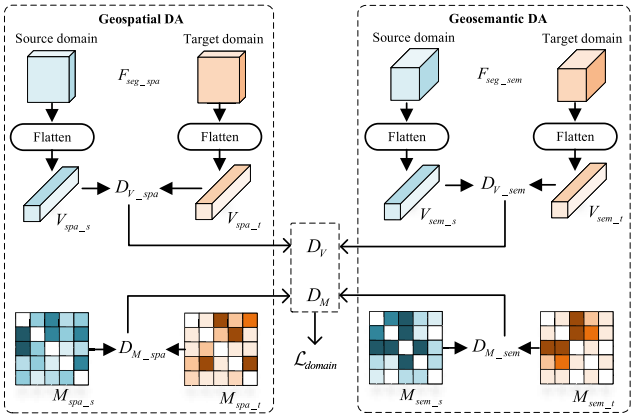


Fig. 6. Illustrations of the proposed GeoSSDA.

where M_{sem} represents the geosemantic correlation matrix, which guides $F_{\text{sem_fusion}}$ to couple the intraclass and decouple the interclass semantic correlations. Finally, a 1×1 convolutional layer is employed for semantic feature embedding. It can be formulated as

$$F_{\text{seg_sem}} = \mathcal{F}_{\text{Conv2d}_{1\times1}}\left(\sigma\left(F_{\text{sem_fusion}} * M_{\text{sem}}\right)\right) \tag{19}$$

where $F_{\text{seg_sem}}$ is fed into the segmentor, which replaces the last layer of the backbone network.

Overall, as depicted in Fig. 5, guided by the geographical context of GeoSpaAI and GeoSemAI, the network is equipped with the essential competence to perceive potential spatial interrelation and semantic correlation.

C. Geospatio-Semantic Domain Adaptation

The efficacy of GeoSpaAI and GeoSemAI has been vividly illustrated in two distinct aspects: spatio-semantic representation and cross-domain adaptation. However, the lack of explicit constraints for domain adaptation leads to a partially misaligned distribution of cross-domain features. In this section, GeoSSDA, a self-training-driven domain adaptation strategy, is presented. As presented in Fig. 6, it establishes hierarchical spatio-semantic consistency within the DSSAL-DA. GeoSSDA mainly consists of two parts: spatio-semantic domain alignment (SSDA) and PCF.

1) Spatial-Semantic Domain Alignment: In DSSAL-DA, there exist two pivotal types of features, which are fusion feature representation V and the correlation matrix M. Specifically, V is divided into the spatial feature V_{spa} and semantic feature V_{sem}, which are produced by flattening operation of $F_{\text{seg_spa}}$ and $F_{\text{seg_sem}}$, respectively. Consequently, these features are deemed indispensable for domain adaptation to alleviate the cross-domain sensitivity of space and semantics. Regarding V, to guarantee its cross-domain capability while avoiding redundant computations, the Euclidean distance calculation is employed to quantify the domain differences between features

$$D_V = \left\|V_{\text{spa_}s} - V_{\text{spa_}t}\right\| + \left\|V_{\text{sem_}s} - V_{\text{sem_}t}\right\|. \tag{20}$$

On the other hand, M represents the absolute approximate distribution of classes, incorporating the deep spatial interrelation M_{spa} and semantic correlation M_{sem}. As a result, the cross-domain differences can be depicted intuitively, and it is considered essential for M to engage in the domain invariance learning. A standard cosine similarity measurement method, denoted as sim(), is utilized for the purpose of learning domain invariance with respect to M

$$D_M = \text{sim}\left(M_{\text{spa_}s}, M_{\text{spa_}t}\right) + \text{sim}\left(M_{\text{sem_}s}, M_{\text{sem_}t}\right). \tag{21}$$

The domain-adaptive strategy mentioned above combined with geospatial prior knowledge, which achieved remarkable performance as described in Section IV-C3.

2) Pseudo-Label Consistency Filter: Typically, self-training methods devise either static or dynamic probability thresholds in an effort to prevent negative transfer caused by misclassification. However, recent pseudo-label updating methods fail to consider the inherent interclass spatial distribution of RSIs, leading to the generation of incomplete and inaccurate pseudo-labels. In the proposed PCF, under the guidance of spatial-semantic correlation, the pseudo-label is derived, which boundary exactitude and central intactness approach GT.

First, moving exponential average (EMA) teacher network [49] without gradient updating is leveraged to generate pseudo-labels, which is formulated as

$$\phi_{t+1} \leftarrow \alpha\phi_t + (1-\alpha)\theta_t \tag{22}$$

where α is the smoothing factor. It weights both student network f_θ and teacher network g_ϕ.

Then, M_{spa} and M_{sem} are jointly employed to facilitate the probability enhancement of pixel-wise pseudo-labels

$$\hat{P}_i = \frac{1}{C}\sum_{c}^{C} P_i \cdot M_{\text{sem}}^{ic} \cdot M_{\text{spa}}^{ic} \tag{23}$$

where $\hat{P}_i$ represents the pixel-wise prediction probability of P_i under the geospatio-semantic constraint. Subsequently, P_i and $\hat{P}_i$ are closely combined to construct the complete pseudo-label consistency representation

$$P_i = \frac{1}{2}\left(P_i + \hat{P}_i\right). \tag{24}$$

Consequently, the updated pseudo-labels encompass a broader expanse of regions, featuring elevated probabilities

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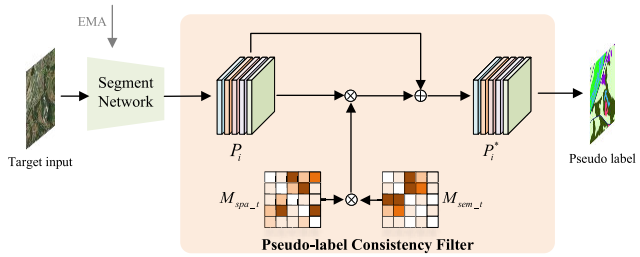


Fig. 7. Illustrations of the proposed PCF.

and enhanced predictions. Simultaneously, with the aid of geospatio-semantic knowledge, the pseudo-labels evolve into a more comprehensive structure under geospatio-semantic guidance, as shown in Fig. 7. Meanwhile, with the assistance of the geospatio-semantic knowledge, the structure of the pseudo-labels becomes more comprehensive.

D. Loss Function

The optimization of the entire model is composed of three loss functions, which include the full-supervision loss in the source domain $\mathcal{L}_s$, domain-adaptive loss $\mathcal{L}_{\text{domain}}$, and self-supervised loss $\mathcal{L}_{\text{pseudo}}$ of pseudo-labels in the target domain

$$\mathcal{L}_s = -\sum_i \mathcal{X}_s^i \log \widetilde{\mathcal{Y}_s^i} \tag{25}$$

$$\mathcal{L}_{\text{domain}} = D_V + D_M \tag{26}$$

$$\mathcal{L}_{\text{pseudo}} = -\sum_i \mathcal{X}_t^i \log \widetilde{\mathcal{Y}_t^i}. \tag{27}$$

Finally, the optimization process of the integrated DSSAL with domain adaptation is described as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_s + \mathcal{L}_{\text{pseudo}} + \lambda \mathcal{L}_{\text{domain}}. \tag{28}$$

It is noted that only the end-to-end $\mathcal{L}_s$ is utilized in the single DSSAL process without domain adaptation.

IV. Experiment and Discussion

A. Datasets and Metrics

1) Datasets: In our experiment, three datasets are employed to validate the performance of the method in diverse cross-domain scenarios. These datasets include two public datasets (ISPRS Potsdam and LoveDA) and a private dataset (Arcmap SX48).

1) The ISPRS Potsdam dataset [50] comprises 38 RSIs of urban areas within the Potsdam region of Germany. The Potsdam region is notable for its extensive clusters of buildings, narrow thoroughfares, and densely packed settlement structures. Each of these images has a pixel dimension of 6000 × 6000 and a spatial resolution of 5 cm and is accompanied by corresponding imagery that has been labeled with six LULC classes, which are classified as impervious surfaces, buildings, low vegetation, trees, cars, and clutter. In the UDA task, the performance of algorithms across different sensors is validated by employing RGB and IRRG cross-domain modalities that cover the same region.
2) The LoveDA dataset [51], provided by Wuhan University, consists of 5987 RGB RS patches. Each of these images has a pixel size of 1024 × 1024 and a spatial resolution of 0.3 m. They depict urban and rural scenes in cities, including Nanjing, Changzhou, and Wuhan in China. In the context of cross-domain urban and rural scenes, seven LULC classes are included, which consist of the background, buildings, road, water, barren, forests, and agriculture.
3) RSIs in the Arcmap SX48 dataset [52] are all derived from the Arcmap platform. Representative scenes from the central basin and southern hilly regions of Shaanxi Province have been selected. Evidently, there are substantial geographical disparities between the basin and hilly regions, especially with regard to vegetation density, water body diversity, village distribution patterns, and urban structural characteristics. This dataset comprises eight RSIs, each having a spatial resolution of 4.8 m and an approximate size of 12 000 × 12 000 pixels. In this dataset, 12 typical classes are fine-grained and geo-expertly annotated, specifically cloud, farmland, grassland, tree, village, city, school, factory, greenhouse, ravine, road, and water. As shown in Fig. 8, compared to the other two public datasets, the Arcmap SX48 dataset has more abundant and more complex land feature classes, which leads to a more challenging domain adaptation for the algorithm.

For domain adaptation, as depicted in Fig. 8, domain discrepancy is pronounced in the statistical characteristics of the different datasets. It illustrates the significance of class-level semantic diversity and the difference of cross-domain land cover distribution.

2) Metrics: In this article, the intersection over union (IoU) is employed to validate the comprehensive performance of category segmentation results and determination. It is used to accurately assess the degree of similarity between the model's prediction results and the GT annotations. IoU measures the overlap between the prediction and the GT by calculating the ratio of the intersection of these two sets to their union. Its mathematical expression is

$$\text{IoU}_i = \frac{M_{ii}}{\sum_{j=0}^{C-1} M_{ij} + \sum_{j=0}^{C-1} M_{ji} - M_{ii}} \tag{29}$$

where M_{ij} indicates the number of examples that belong to class i being predicted as class j.

Furthermore, the IoU is calculated separately, yielding the mean IoU (mIoU), which is employed to assess the overall performance of the algorithm in both semantic segmentation task and domain adaptation of cross-domain scenarios.

B. Implementation Details

For DSSAL, four distinct backbone networks, consisting of CNN-based, ConvNeXt-based, Transformer-based, and Mamba-based. They are employed to examine the scalability and robustness of the proposed method. Specifically, for

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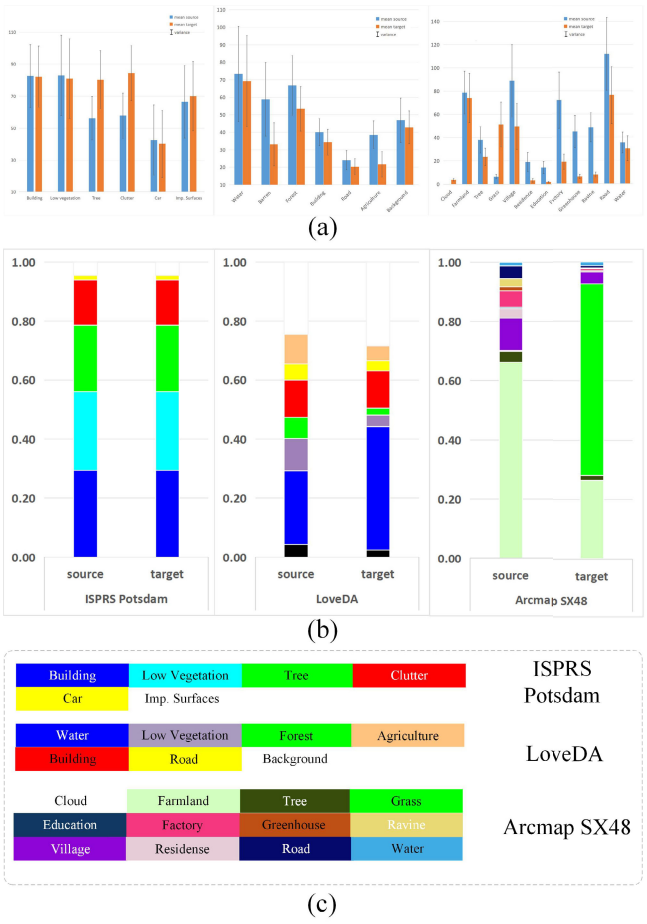


Fig. 8. Domain discrepancy is pronounced in the statistical characteristics of the three different datasets. (a) Significance of class-level semantic diversity. (b) Difference of cross-domain land cover distribution. (c) Colorbar of the utilized three datasets. Specifically, in the ISPRS Potsdam dataset, there are different sensors under RGB/IRRG spectral bands for the same scenario and class. In the LoveDA dataset, it exhibits different city styles with the same seven basic land use classes. The Arcmap SX48 dataset presents different and more fine-grained classes, along with an unbalanced distribution within the same 12 land cover classes. These all reflect that abundant semantic information is typically indicated by intraclass feature discrepancy and interclass feature distinctiveness.

the CNN-based approach, the DeepLabV3+ [53] structure with the complete ResNet-101 backbone is utilized. In the case of the ConvNeXt-based and Transformer-based methods, the ConvNeXt-Tiny [54] and Swin-Tiny [55] architectures are adopted, respectively. To align with the latest advancements in the field, the VMamba-Tiny [56] is employed to implement the geographical correlation guidance. Furthermore, nine SOTA RSISS networks in UDA, namely AdaptSegNet [16], FADA [57], DCA [58], ProCA [25], CLA-UDA [39], MBATA [22], MemoryAdaptNet [46], MIC [19], and UemDA [42], are employed to examine the scalability and robustness of the proposed DSSAL-DA.

With ImageNet-1K pretraining serving as the foundation, the initial learning rates for the aforementioned four architectures are set to 5×10^{-3}, 5×10^{-5}, 5×10^{-5}, and 5×10^{-6}. For the DeepLabV3+ architecture, it is trained for 100 epochs using the stochastic gradient descent (SGD) optimizer along with the poly learning rate update strategy. In contrast, for the remaining three architectures, the configurations adopt the AdamW optimizer with a weight decay of 1×10^{-4} and the cosine annealing learning rate update strategy, and the training process is set to 80 epochs. In other respects, α and λ are set to 0.996 and 0.001, respectively. All of the experiments described above are executed on the Ubuntu 18.04 LTS platform, leveraging a single NVIDIA GeForce RTX 3090 24G GPU.

C. Ablation Study

In this section, we conduct comprehensive experiments and engage in in-depth discussions regarding the cross-domain semantic segmentation effectiveness of the proposed DSSAL and DSSAL-DA across three datasets. The content primarily includes the experimental results obtained with different backbones, discussions on the domain adaptation capabilities of DSSAL, and the outcomes achieved by different modules of DSSAL-DA.

1) Discussion on Different Backbones: Table I illustrates the evaluation metrics for the ISPRS Potsdam dataset. Among the various approaches, CNN-based methods showed comprehensive improvement and outperformed others under the DSSAL structure, with the mIoU increasing by 0.0558. Although the Swin-, ConvNeXt-, and VMamba-based methods did not exhibit substantial segmentation improvements in certain classes, they demonstrated notable enhancements in classes characterized by distinct spatial structures, such as the building and the clutter. Moreover, they maintained the best overall segmentation results, with the mIoU increasing by 0.0085, 0.0122, and 0.0117 compared to the cases without using DSSAL. This proves that DSSAL has a more remarkable sensitivity to the spatial correlations between classes. Additionally, when compared to the scenarios without the DSSAL, these three methods maintained superior overall segmentation with the mIoU increasing by 0.0085, 0.0122, and 0.0117. The findings provide evidence that DSSAL possesses a more pronounced sensitivity to the interclass geospatial interrelations. Furthermore, (1, 6, 256, 256) is used as input shape to measure the parameters and computational costs of disparate models. Due to the different feature dimensions of each block in diverse backbones, there will be certain discrepancies in parameters and computational costs. However, DSSAL has achieved significant superiority in both quantitative and qualitative results by increasing only a few parameters in general.

In addition, Fig. 9 showcases the visualization results of eight randomly selected groups that cover various scenarios with the proposed DSSAL. The enhancements brought about by DSSAL in terms of class integrity and classification accuracy are quite evident. Specifically, in the cross-domain scenarios of the ISPRS Potsdam dataset, the interclass spatial relationships remain consistent, whereas the intraclass semantic disparities are remarkably pronounced. Particularly in areas where the boundaries between regions are ambiguous and the class structure information is confounding, as demonstrated in the visualization result of the seventh column in Fig. 9, the accuracy of the boundary of the blue building with a distinct structure was enhanced by DSSAL. In the eighth column, owing to the combined influences of GeoSpaAI and GeoSeAI, the interclass confusion among different backbones in

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TABLE I
EXPERIMENTAL ABLATION RESULTS UNDER THE DIFFERENT BACKBONE NETWORKS. THE BEST RESULTS ARE IN BOLD

Method	DSSAL	Params(M)	GFLOPs	Class						mIoU
				surface	building	low vegs	tree	car	clutter	
ResNet-101	×	58.75	15.18	0.6443	0.7254	0.3201	0.5646	0.7466	0.1845	0.5309
	✓	74.28	36.07	**0.6999**	**0.7808**	**0.4375**	**0.5992**	**0.7792**	**0.2238**	**0.5867**
Swin-T	×	63.71	85.24	**0.7098**	0.7191	0.3781	**0.6154**	**0.8022**	0.2992	0.5873
	✓	65.92	88.05	0.7062	**0.7523**	**0.4037**	0.6137	0.7812	**0.3174**	**0.5958**
ConvNeXt-T	×	63.98	83.94	0.6595	0.7582	0.3241	**0.6008**	0.785	0.2212	0.5581
	✓	66.18	86.71	**0.6704**	**0.7708**	**0.3697**	0.5963	**0.7902**	**0.2246**	**0.5703**
VMamba-T	×	40.10	20.98	0.7374	0.818	0.4723	**0.6454**	**0.7936**	0.2759	0.6238
	✓	43.68	22.70	**0.7407**	**0.8449**	**0.5091**	0.6397	0.7899	**0.2884**	**0.6355**

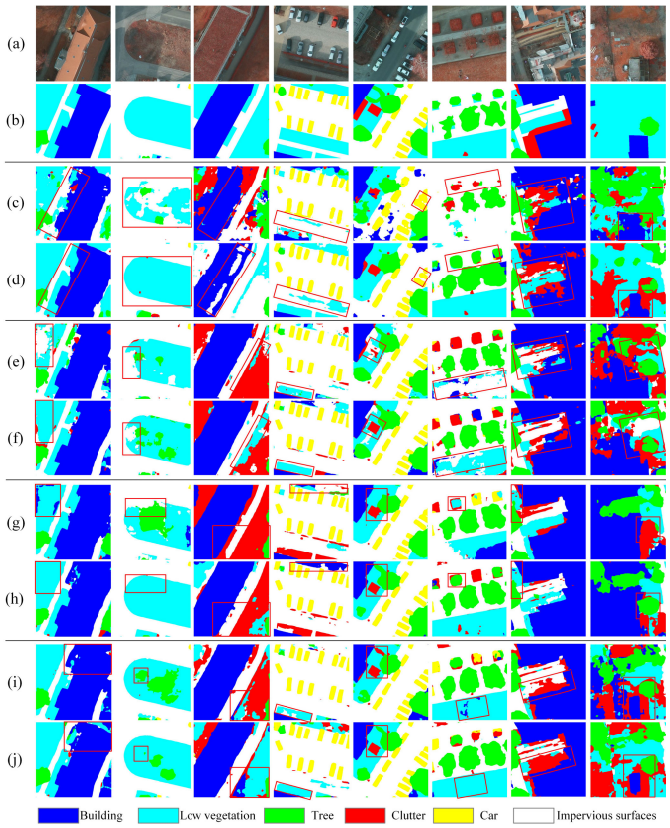


Fig. 9. Visualization results of different backbone networks in the ISPRS Potsdam (RGB → IRRG) task. (a) RGB images. (b) GTs. (c) Results of ResNet-101. (d) Results of ResNet-101 w/ DSSAL. (e) Results of Swin-T. (f) Results of Swin-T w/ DSSAL. (g) Results of ConvNeXt-T. (h) Results of ConvNeXt-T w/ DSSAL. (i) Results of VMamba-T. (j) Results of VMamba-T w/ DSSAL. The significantly improved regions are highlighted in red.

complex scenarios was alleviated, bringing the segmentation results closer to the GT. DSSAL guarantees the continuity of the impervious surface and the completeness of the car, thereby reflecting the beneficial constraints guided by geospatial knowledge. On the other hand, in regions characterized by abundant geosemantic information, such as the low vegetation and the building, the proposed DSSAL demonstrates remarkable consistency in mitigating the intraclass confusion.

2) Discussion on Domain Adaptation of DSSAL: To verify and confirm the latent domain adaptation capacity of the DSSAL methodology, a series of visual outcomes are shown in Fig. 10. The segmentation outcomes of various base models (Deeplabv3+, ConvNeXt-T, Swin-T, and VMamba-T) with and without DSSAL are illustrated. Without DSSAL, the base models exhibit certain misclassifications and less precise segmentations. For example, in the segmentation results of Deeplabv3+, some low vegetation areas are misidentified as other classes. Conversely, when DSSAL is integrated, the segmentation boundaries become more distinct, and the misclassification rates decline. For instance, in the Deeplabv3+ with DSSAL results, the classification of low vegetation areas aligns more closely with the GT. In Fig. 10(a), the segmentation results of various base models (Deeplabv3+, ConvNeXt-T, Swin-T, and VMamba-T) both with and without the integration of DSSAL are depicted. In the absence of DSSAL, these base models display certain misclassifications and less accurate segmentations. For example, in the segmentation outcomes of Deeplabv3+, some regions of low vegetation are misclassified as other classes. On the contrary, when DSSAL is incorporated, the segmentation boundaries become significantly pronounced, and the misclassification rates decrease as shown in highlighted regions. For instance, in the segmentation results of Deeplabv3+ with DSSAL, the classification of low vegetation regions is substantially consistent with the GT. Fig. 10(b) illustrates the segmentation probability distributions. Base models without the integration of DSSAL exhibit more dispersed and less assured probability distributions in certain regions, which reflects the ambiguity in classification. In comparison, models incorporating the DSSAL feature more concentrated probability distributions, indicating that DSSAL empowers the models to make more confident and precise classification decisions. Fig. 10(c) demonstrates the accuracy of pixel-level segmentation. Black pixels are indicative of incorrect segmentation, whereas white pixels signify correct segmentation. Evidently, models incorporating DSSAL exhibit a more limited number of uncorrect black pixels, suggesting a higher degree of pixel-level segmentation accuracy when compared to base models lacking DSSAL. Fig. 10(d) presents the t-SNE visualizations of various intraclasses and interclasses. The class clusters of models integrated with DSSAL are more compact and distinctly delineated, which indicates that DSSAL is capable of effectively differentiating between different classes within the feature space. Conversely, the class clusters of base models without the incorporation of DSSAL are more scattered, implying a lower level of effective class discrimination. In conclusion, these visual results comprehensively demonstrate the remarkable domain adaptation of DSSAL. It enables DSSAL to

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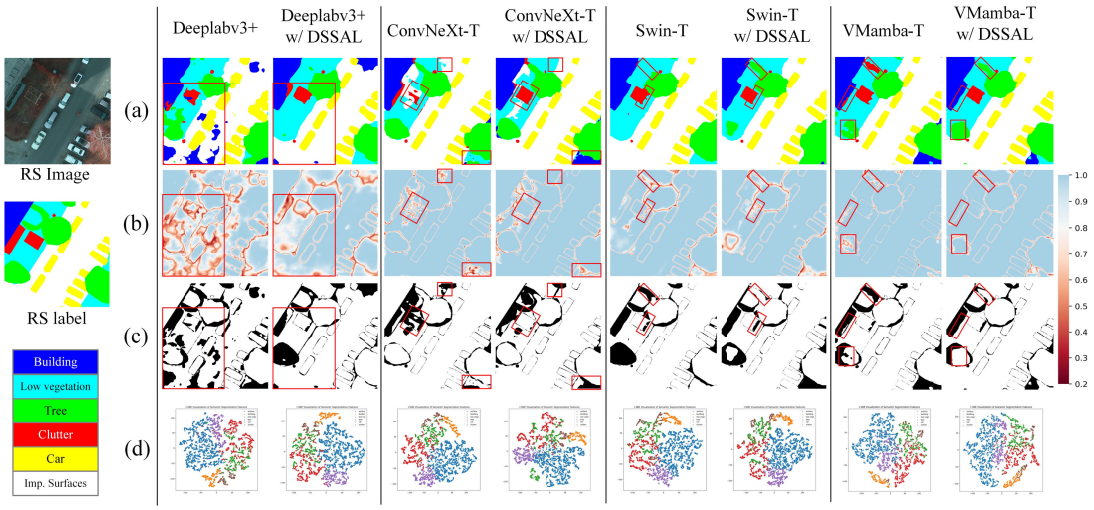


Fig. 10. Visualization results of the ISPRS Potsdam (IRRG → RGB) task. (a) Results of different models. (b) Predict the probability of (a). (c) Error map of (a). (d) Class-wise feature distribution based on t-SNE. The significantly improved regions are highlighted in red.

TABLE II

EXPERIMENTAL ABLATION RESULTS UNDER THE DIFFERENT MODULES IN THE ISPRS POTSDAM DATASET. THE BEST RESULTS ARE IN BOLD

Method	Class						mIoU
	surface	building	low vegs	tree	car	clutter	
Source Only (Deeplab-based)	0.6443	0.7254	0.3201	0.5646	0.7466	0.1845	0.5309
w/ GeoSemAI	0.6704	0.7489	0.3698	0.5388	0.7453	0.1846	0.5430
w/ GeoSpaAI	0.7078	0.7442	0.4175	0.5786	0.7878	0.2139	0.5750
w/ GeoSemAI & GeoSpaAI	0.6999	0.7808	0.4375	0.5992	0.7792	0.2238	0.5867
Based GeoSemAI & GeoSpaAI							
w/ GeoSSDA	0.7139	0.8235	0.5556	0.6158	0.7723	0.2652	0.6244
w/ PCF	0.6807	0.7972	0.4610	0.5873	0.7885	0.2472	0.5937
DSSAL_DA	**0.7348**	**0.8599**	**0.6279**	**0.6324**	**0.8002**	**0.3124**	**0.6613**

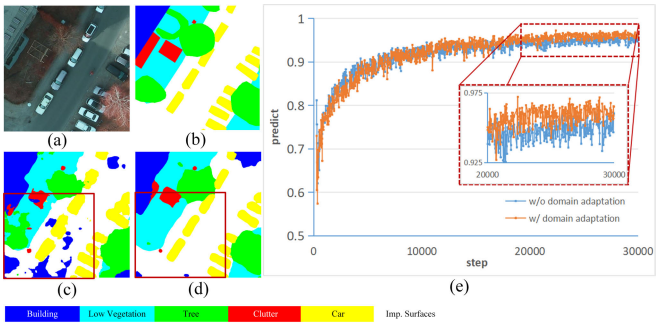


Fig. 11. Illustration of the latent restrictions of domain adaptation. (a) RSI. (b) GT of RSI. (c) Prediction of DeepLabV3+. (d) Prediction of DeepLabV3+ with GeoSSDA. (e) Despite the notable enhancement in segmentation results, there is no substantial increase in the prediction probability. It implies that there are still a number of ambiguously uncertain pixels present during the pseudo-label training.

substantially improve the segmentation performance of various base models in the context of RSISS.

3) Discussion on Different Modules of DSSAL-DA: To evaluate the effectiveness of individual components in our proposed DSSAL-DA for UDA semantic segmentation of RSIs, we conducted ablation experiments on the ISPRS Potsdam dataset. Table II illustrates the objective evaluation outcomes. The "Source Only (Deeplab-based)" approach is employed as the baseline. When the GeoSemAI module is utilized in isolation, a noticeable improvement is observed for the majority of classes. For example, the mIoU of the surface increases from 0.6443 to 0.6704. The addition of the GeoSpaAI module likewise results in performance enhancements. Specifically, the mIoU of the car elevates from 0.7466 to 0.7878. When combining GeoSemAI and GeoSpaAI, further improvements are observed, demonstrating the complementary nature of these two modules. It is worth noting that domain loss effectively mitigates the overfitting problem in the source domain, as shown in Fig.11. Although the improvement in classification probability is not substantial, yet the segmentation results exhibit the critical role of GeoSSDA, as exhibited in Fig. 11(c) and (d). On the other hand, when PCF is employed alone, the mIoU is 0.0307 lower than GeoSSDA, indicating that domain alignment plays a crucial role in the adaptive process. Incorporating the domain-adaptation method SSDA and the pseudo-label optimization method PCF (DSSAL-DA) results in even higher mIoU values across all classes, indicating the effectiveness of these additional components in our overall framework. Upon integrating the GeoSemAI and GeoSpaAI modules, additional enhancements are detected, which vividly demonstrates the complementary characteristics of these two modules. The integration of the domain adaptation approach

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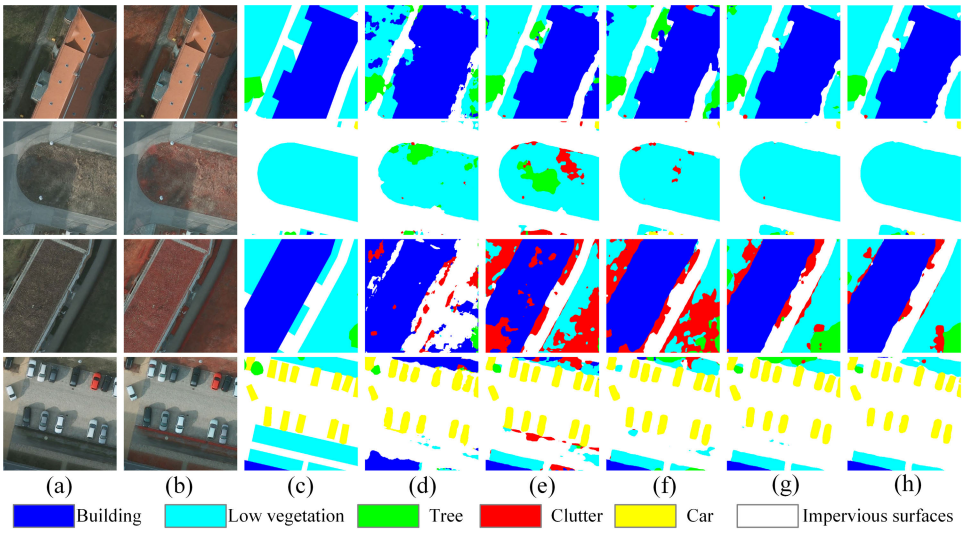


Fig. 12. Visualization results of different modules in the ISPRS Potsdam (IRRG → RGB) task. (a) RGB images. (b) IRRG images. (c) Labels of (b). (d) Results of w/ GeoSemAI. (e) Results of w/ GeoSpaAI. (f) Results of w/ GeoSemAI and GeoSpsAI. (g) Results of w/ SSDA. (h) Results of w/ PCF.

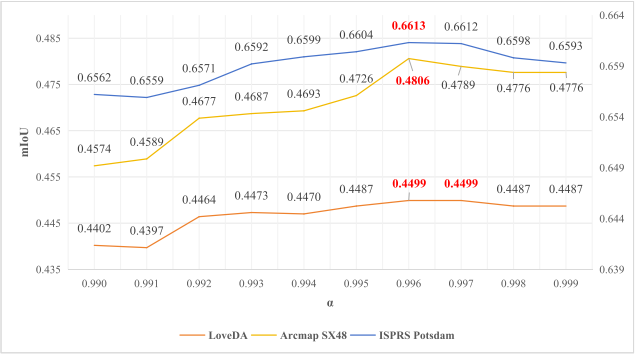


Fig. 13. Illustration of different smoothing factors α of EMA.

GeoSSDA and the PCF generates even superior mIoU values across all classes. This outcome effectively highlights the efficacy of these supplementary components within our comprehensive framework.

The visual outcomes presented in Fig. 12 further validate our findings. Fig. 12(a) displays the original RSIs, whereas Fig. 12(b) showcases the corresponding GT labels. Fig. 12(c) illustrates the segmentation results of the source-only approach, which exhibits several misclassifications and less precise boundaries. Fig. 12(d) and (e) depicts the results obtained by adding GeoSpaAI and GeoSemAI individually. These results demonstrate enhanced segmentation accuracy when compared to the source-only method. Fig. 12(f), which incorporates both GeoSpaAI and GeoSemAI, yields more accurate segmentations. Fig. 12(g), featuring domain adaptation, and Fig. 12(h), with pseudo-label optimization, further enhance the results, minimizing misclassifications and improving the overall visual quality of the segmentation. The ablation experiments comprehensively demonstrate the significance and efficacy of each component within our proposed DSSAL-DA for RSISS. Moreover, different smoothing factors of EMA are utilized to evaluate the domain adaptation performance of the proposed method as shown in Fig. 13. Through experimental evaluation, it is found that the best result is achieved when the value is set at 0.996.

D. Discussion on Comparison and Analysis With SOTA Methods

In this section, three distinct UDA tasks, as depicted in Fig. 1, are selected to compare the proposed DSSAL-DA to the SOTA methods.

1) ISPRS Potsdam: Table III illustrates the quantitative evaluation outcomes, presenting the IoU values for each individual class and the mIoU across all classes. The proposed DSSAL-DA method surpasses numerous existing domain adaptation approaches in the majority of classes. In the case of the surface, DSSAL-DA attains an IoU value of 0.7348, which exceeds the IoU values of most of the compared methods. Regarding the building, DSSAL-DA achieves an IoU of 0.8599, significantly outperforming several other approaches. In terms of the mIoU, DSSAL-DA also demonstrates outstanding performance. Specifically, it reaches an mIoU of 0.6613 when utilizing the ResNet-101 and 0.6831 when employing the MiT-B5 backbone. These mIoU values rank among the highest within the comparison. These visual and quantitative results comprehensively demonstrate that our DSSAL-DA approach possesses a significant competitive edge over other domain adaptation methodologies in the context of RSISS on the ISPRS Potsdam dataset, providing more accurate and reliable segmentation results.

Fig. 14 elucidates the visualization outcomes. Fig. 14(a) portrays the RSIs, while the second column, labeled as Fig. 14(b), exhibits their corresponding GT. Fig. 14(c)–(m) presents the segmentation results of diverse methods that employ the ResNet-101 backbone. These methods are arranged in a top-to-bottom order as depicted in Fig. 14. Visually, when contrasted with other methodologies, the segmentation results of DSSAL display more precise boundaries and a reduced number of misclassifications. For instance, in the segmentation of building and low vegetation areas, DSSAL-DA is more adept at differentiating these classes from the background and other objects, thereby yielding a result that is more highly congruent with the GT.

2) LoveDA: We perform a comparison of the proposed DSSAL-DA with other SOTA domain adaptation techniques

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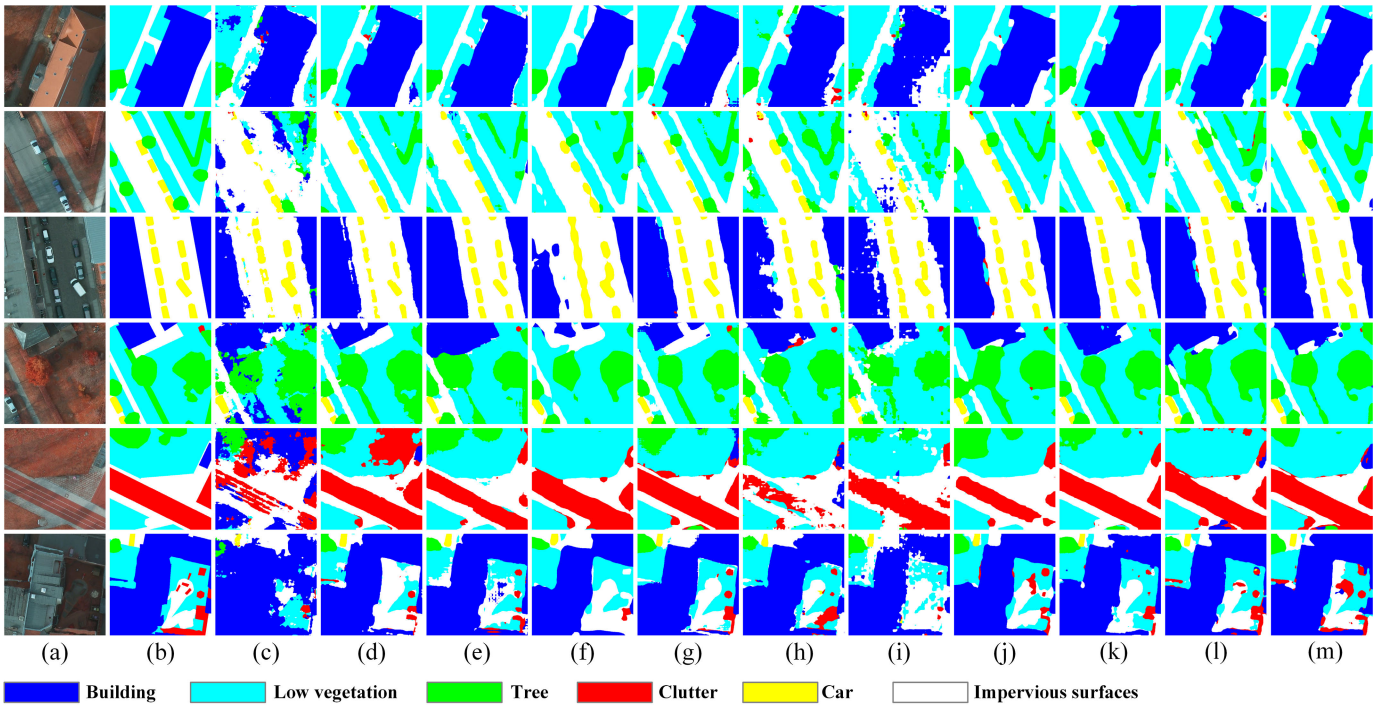


Fig. 14. Visualization results of the ISPRS Potsdam (IRRG → RGB) task. (a) Input images. (b) Label of input images. (c) Results of Source Only. (d) Results of AdaptSegNet. (e) Results of FADA. (f) Results of DCA. (g) Results of ProCA. (h) Results of CLA-UDA. (i) Results of MBATA. (j) Results of MemoryAdaptNet. (k) Results of MIC. (l) Results of UemDA. (m) Results of our proposed DSSAL-DA.

TABLE III

EXPERIMENTAL RESULTS UNDER THE VARIATIONS IN BOTH GEOGRAPHIC LOCATIONS AND IMAGING MODE: QUANTITATIVE COMPARISON WITH THE SOTA ON THE ADAPTATION POTSDAM RGB TO IRRG. THE BEST RESULTS ARE IN BOLD

Task	Backbone	method	Class						mIoU
			surface	building	low vegs	tree	car	clutter	
RGB ↓ IRRG	ResNet-101	Source Only	0.5673	0.5829	0.2620	0.4820	0.5632	0.1343	0.4320
		AdaptSegNet	0.6563	0.7982	0.4649	0.5978	0.7239	0.2503	0.5819
		FADA	0.6911	0.8191	0.5670	0.6279	0.7521	0.2583	0.6193
		DCA	0.6816	0.8046	0.6292	0.6181	0.7303	0.2303	0.6157
		ProCA	0.7223	0.8388	0.5721	0.6326	0.7651	0.3014	0.6387
		CLA-UDA	0.6413	0.7825	0.5800	0.5833	0.7160	0.1975	0.5834
		MBATA	0.6301	0.7644	0.5063	0.5722	0.7266	0.2878	0.5812
		MemoryAdaptNet	0.7191	**0.8649**	0.6250	0.6385	0.7635	0.2905	0.6503
		MIC	**0.7394**	0.8442	**0.6353**	**0.6414**	0.7647	**0.3309**	0.6593
		UemDA	0.7032	0.8145	0.6296	0.5979	0.7520	0.2898	0.6312
		Ours	0.7348	0.8599	0.6279	0.6324	**0.8002**	0.3124	**0.6613**
	MiT-B5	MIC	0.7347	0.8619	0.6104	**0.6559**	0.7983	0.3353	0.6661
		Ours	**0.7581**	**0.8763**	**0.6557**	0.6543	**0.8061**	**0.3479**	**0.6831**

on the LoveDA dataset. The experimental results in the Urban→Rural domain adaptation task demonstrate the superiority of our method across multiple semantic categories. As shown in Table IV, the proposed DSSAL-DA achieves a remarkable IoU of 0.4274 and 0.3907 in building and barren. Notably, DSSAL-DA exhibits balanced performance with an overall mIoU of 0.4499 and 0.4884 under ResNet-101 and MIT-B5 backbone, respectively, outperforming the second-best MIC. While domain alignment methods like ProCA show moderate improvements in urban features (e.g., 0.4215 IoU in the building), they underperform in rural-specific categories like Barren (0.2415). The results highlight our method's enhanced capability in addressing rural domain shifts through effective feature alignment and context-aware adaptation. In particular, the 0.6314 IoU in the water and 0.4470 in the road with MIT-B5 further confirm its robustness to diverse rural topographies.

As shown in Fig. 15, our proposed DSSAL-DA (last column) outperforms other methods in cross-domain semantic segmentation of RSIs. For classes like water and road, it achieves sharper boundary delineation, reducing misclassifications evident in other approaches. In segmenting building and agriculture, our method ensures higher integrity—avoiding the fragmented results seen in methods such as CLA-UDA

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TABLE IV

EXPERIMENTAL RESULTS UNDER THE VARIATIONS IN BOTH GEOGRAPHIC LOCATIONS AND IMAGING MODE: QUANTITATIVE COMPARISON WITH THE SOTA ON THE ADAPTATION LoveDA URBAN TO RURAL. THE BEST RESULTS ARE IN BOLD

Task	Backbone	method	Class							mIoU
			background	building	road	water	barren	forest	agriculture	
Urban → Rural	ResNet-101	Source Only	0.4615	0.3080	0.2329	0.3500	0.0529	0.0793	0.3277	0.2589
		AdaptSegNet	0.3980	0.3605	0.3102	0.3461	0.1681	0.2016	0.1548	0.2770
		FADA	0.4825	0.3835	0.2804	0.4694	0.1651	0.2098	0.4043	0.3421
		DCA	0.4835	0.3127	0.3843	0.4851	0.1446	0.2267	**0.6581**	0.3850
		ProCA	0.4963	0.4215	0.3312	0.5081	0.2415	0.2239	0.3618	0.3692
		CLA-UDA	0.3889	0.4069	0.2500	**0.5563**	0.1344	0.2821	0.5203	0.3627
		MBATA	**0.5280**	0.3944	0.3185	0.3473	0.1864	0.2129	0.1973	0.3121
		MemoryAdaptNet	0.4508	0.4234	**0.4008**	0.4325	0.2343	0.2652	0.3046	0.3588
		MIC	0.4834	0.3745	0.2991	0.5207	0.1734	**0.3395**	0.3384	0.3613
		UemDA	0.4903	**0.4443**	0.3739	0.4922	0.1495	0.2714	0.4728	0.3849
		Ours	0.5047	0.4274	0.3608	0.5402	**0.3907**	0.2681	0.6575	**0.4499**
	MiT-B5	MIC	0.4469	0.4656	0.4036	0.6262	0.4039	0.3820	0.4357	0.4520
		Ours	**0.4815**	**0.4712**	**0.4470**	**0.6314**	**0.4071**	**0.3832**	**0.5976**	**0.4884**

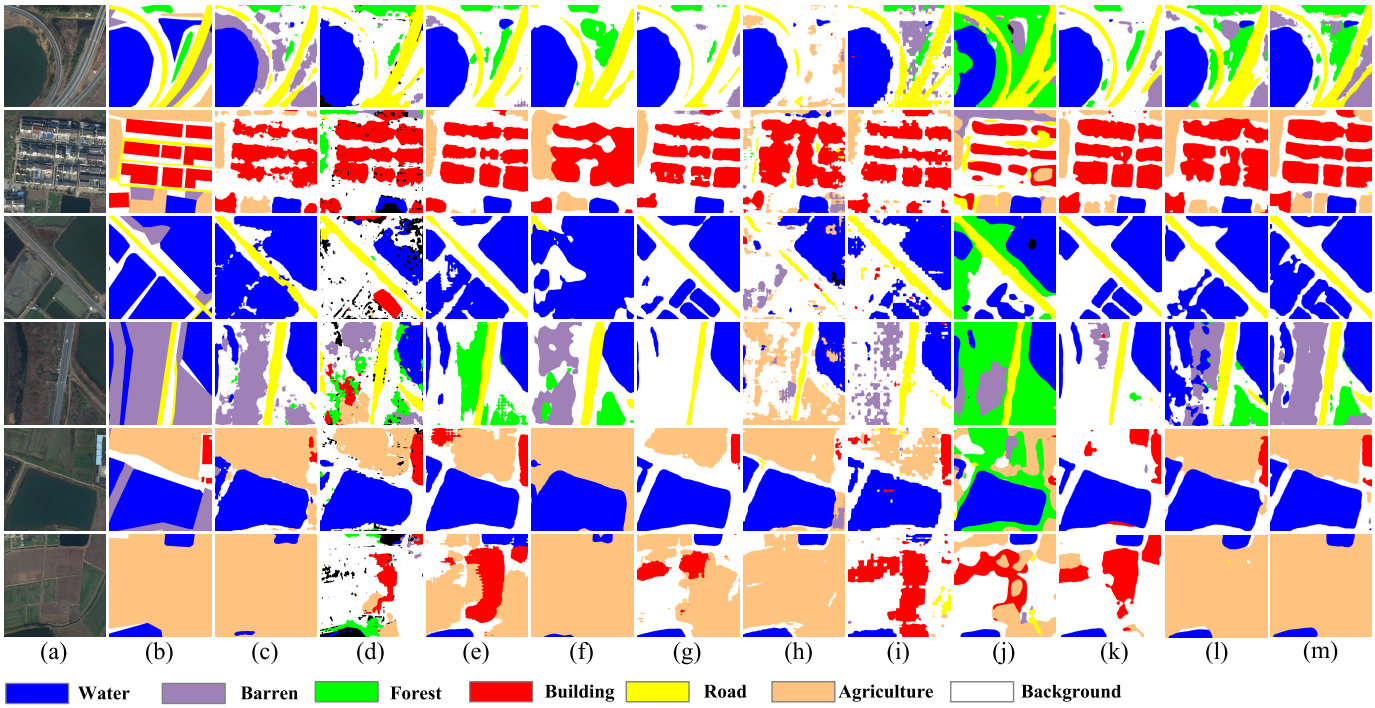


Fig. 15. Visualization results of the LoveDA (Urban → Rural) task. (a) Input images. (b) Label of input images. (c) Results of Source Only. (d) Results of AdaptSegNet. (e) Results of FADA. (f) Results of DCA. (g) Results of ProCA. (h) Results of CLA-UDA. (i) Results of MBATA. (j) Results of MemoryAdaptNet. (k) Results of MIC. (l) Results of UemDA. (m) Results of our proposed DSSAL-DA.

or FADA. Crucially, it adeptly addresses urban–rural domain shifts, maintaining structural consistency and enhancing class discriminability. Such performance highlights its superior adaptability and precision in handling domain-adaptive semantic segmentation.

3) Arcmap SX48: The experimental results in Table V demonstrate the superior performance of the proposed DSSAL-DA compared to SOTA approaches in cross-domain adaptation from Arcmap SX48 Basin to Hills. When employing the ResNet-101 as the backbone, DSSAL-DA, respectively, achieves leading mIoU and mIoU* of 0.4606 and 0.4806, which are superior to the second-best method 0.0314 and 0.0559. Particularly noteworthy are the significant improvements in challenging classes, including farmland, village, factory, and road. With the MIT-B5 backbone, the proposed DSSAL-DA maintains consistent superiority across multiple IoU, especially in vegetation-related classes like farmland (0.7697) and tree (0.4835). The red-highlighted values indicate DSSAL-DA optimal performance in 8 out of 12 classes under cross-domain adaptation scenarios, demonstrating enhanced domain-invariant feature learning capability and spatial integrity. The results validate the effectiveness of our architectural innovations in addressing geographic and

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TABLE V

EXPERIMENTAL RESULTS UNDER THE VARIATIONS IN BOTH GEOGRAPHIC LOCATIONS AND IMAGING MODE: QUANTITATIVE COMPARISON WITH THE SOTA ON THE ADAPTATION ARCMAP SX48 BASIN TO HILLS. THE BEST RESULTS ARE IN BOLD. * INDICATES THE RESULTS OF METHODS WITHOUT CATEGORIES SCHOOL AND GREENHOUSE

Task	Backbone	method	Class												mIoU	mIoU*
			cloud	farmland	grass	tree	village	urban	school	factory	greenhouse	ravine	road	water		
Basin ↓ Hills	ResNet-101	Source Only	0.3316	0.4535	0.1711	0.7697	0.4814	0.3392	0.2111	0.2947	0.3303	0.0887	0.1460	0.3347	0.3293	0.3691
		AdaptSegNet	0.3098	0.4244	0.1776	0.5514	0.4935	0.2358	0.4114	0.3732	**0.5263**	0.0997	0.1954	0.3952	0.3495	0.3507
		FADA	0.4786	0.4699	0.1631	0.7540	0.5014	0.3537	0.3030	0.3962	0.2865	0.1418	0.1553	0.3660	0.3641	0.4042
		DCA	0.0901	0.4237	0.1646	0.8110	0.4351	0.5088	—	0.3819	0.3645	—	0.0915	0.2863	0.3558	0.3548
		ProCA	0.6314	0.4691	0.1967	0.7329	0.4736	0.4572	0.4319	0.3887	0.2697	0.1241	0.1740	0.4460	0.3996	0.4411
		CLA-UDA	**0.6590**	0.4759	0.1884	0.7548	0.4027	0.3281	—	0.2634	—	0.0775	0.1104	0.2888	0.3549	0.3857
		MBATA	0.4571	0.4562	**0.2526**	0.7233	0.4541	0.4207	0.2812	0.2494	0.1712	**0.2919**	0.0773	0.2637	0.3416	0.3727
		MemoryAdaptNet	0.6236	0.3433	0.2307	**0.8243**	0.4959	0.3945	0.6003	0.4079	0.5135	0.2148	0.1266	0.3755	0.4292	0.4247
		MIC	0.4420	0.4301	0.0743	0.6316	0.4845	0.4630	**0.7340**	0.3555	0.3869	0.1139	0.2170	0.4629	0.3996	0.3957
		UemDA	0.3913	0.4173	0.1840	0.7945	0.4531	**0.5093**	0.5798	0.4053	0.3904	0.1872	0.1825	0.4016	0.4080	0.4154
		Ours	0.6268	**0.5142**	0.2445	0.8003	**0.5411**	0.4533	0.4739	**0.4371**	0.5062	0.2222	**0.2350**	**0.4728**	**0.4606**	**0.4806**
	MiT-B5	MIC	0.5592	0.5296	**0.2809**	0.8308	**0.5700**	**0.4547**	0.508	**0.4393**	0.5941	0.2458	0.2279	0.4698	0.4758	0.4847
		Ours	**0.6184**	**0.5328**	0.2390	**0.8361**	0.5691	0.4062	**0.5726**	0.4376	**0.6071**	**0.2759**	**0.2522**	**0.4855**	**0.4860**	**0.4863**

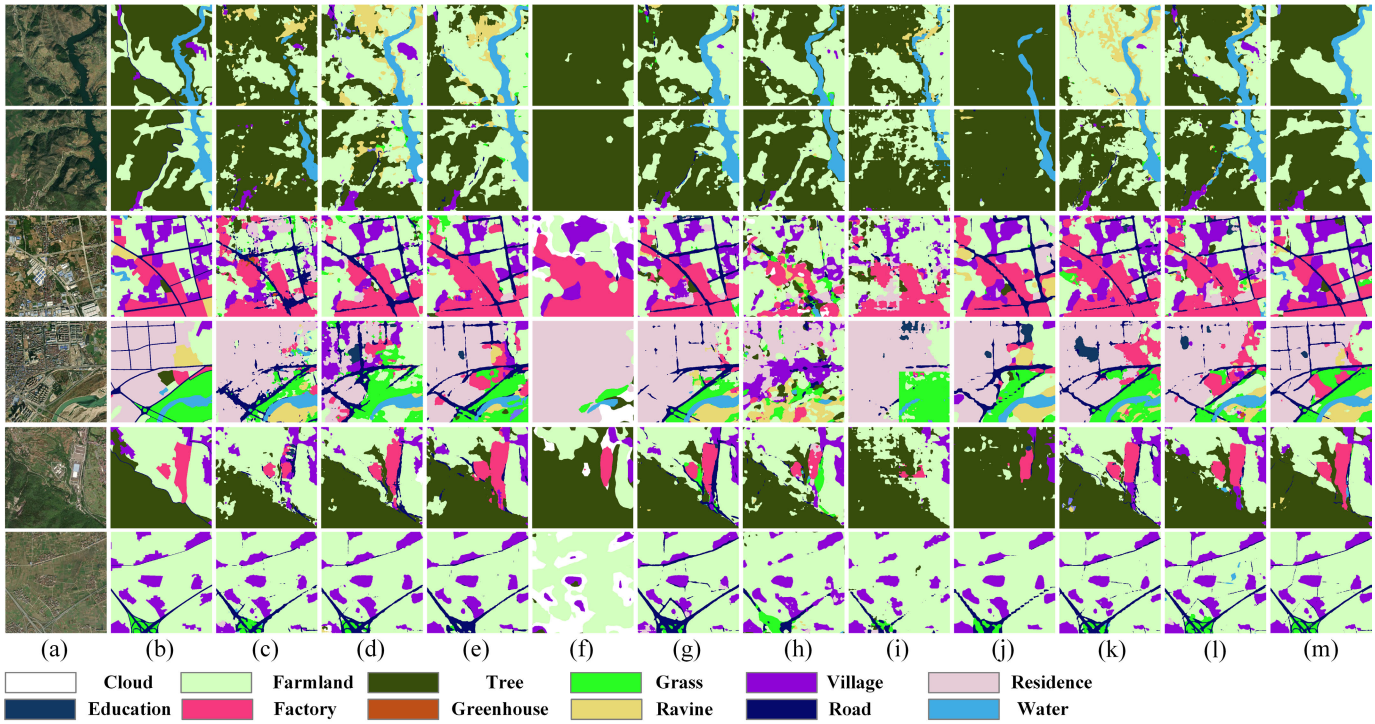


Fig. 16. Visualization results of the Arcmap SX48 (Basin → Hills) task. (a) Input images. (b) Label of input images. (c) Results of Source Only. (d) Results of AdaptSegNet. (e) Results of FADA. (f) Results of DCA. (g) Results of ProCA. (h) Results of CLA-UDA. (i) Results of MBATA. (j) Results of MemoryAdaptNet. (k) Results of MIC. (l) Results of UemDA. (m) Results of our proposed DSSAL-DA.

imaging modality discrepancies compared to existing adversarial (AdaptSegNet and MBATA), prototype-based (ProCA and MemoryAdaptNet), and mask-based (MIC) approaches.

As depicted in Fig. 16, in addition to the initial two columns (input images and GT), each column presents cross-domain semantic segmentation visualization results of different methods. Our proposed DSSAL-DA (final column) outperforms significantly. Compared with the SOTA methods, it exhibits more precise boundary delineation and mitigates misclassifications across classes such as village and road by geospatio-semantic guidance. Importantly, it effectively preserves structural details and enhances class discriminability, demonstrating superior adaptability in cross-domain RSISS. The robustness of its ability to handle domain shifts is underscored by its superior performance in both accuracy and structural detail retention compared to competing approaches.

V. CONCLUSION AND FUTURE WORK

In this article, we propose a novel semantic segmentation method DSSAL in RSIs with geographical correlation guidance. DSSAL is applied to cross-domain remote sensing scenarios and a domain-adaptive RSISS network DSSAL-DA is represented, which addresses the challenges of cross-domain shifts and interclass ambiguity in RSIs. Specifically, the GeoSpaAI captures intraclass spatial continuity and interclass adjacency patterns through connected domain analysis,

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while GeoSemAI leverages class prototypes to disentangle semantic confusion via covariance-based correlation matrices. Furthermore, the proposed GeoSSDA aligns cross-domain spatio-semantic distributions via feature-level alignment and correlation-level constraints. For self-training, PCF mitigates error propagation and enhances pseudo-label reliability. Extensive experiments across three datasets demonstrate that DSSAL and DSSAL-DA outperform the SOTA methods in cross-domain scenarios. Our work underscores the importance of embedding geographical priors into the domain adaptation framework for RSIs. In the future, we will focus our attention on leveraging the inherent a priori properties of real-world objects to improve the shortcomings of merely constructing potential geographical spaces and semantic relationships through pure algorithms.

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